ZCAS University

BACHELOR OF SCIENCE IN NETWORK ENGINEERING

CCS2142 PROGRAMMING IN C

MID SEMESTER TEST

12:30 -15:30

Friday 20th October 2023

TIME ALLOWED: THREE HOURS

INSTRUCTIONS:

1. Section A: ALL questions in this section are COMPULSORY and must be attempted
2. Section B: Answer any **THREE** questions only
3. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: All questions in this section are compulsory and must be answered

QUESTION 1

Wina Bwangu Enterprise has booths across Lusaka. The offer mobile money services. Currently they physically go round to monitor operations. This is proving tedious as more booths are opening in various locations. The marketing manager has been advised to develop a system that can help them monitor all the operations and transactions taking place across all booths across Lusaka. They have hired you as a programmer to help them develop a system. They have been told that C programming language is a high-level language and can be used to develop the application they need. They have some information that C is a general-purpose programming language created by Dennis Ritchie at the Bell Laboratories in 1972.It is a very popular language, despite being old. The main reason for its popularity is because it is a fundamental language in the field of computer science. Therefore,

(a) Briefly explain ANY THREE features of the C programming language (6marks)

(b) Describe a variable in C and its attributes. Give an example by declaring a variable using any data type. (6marks)

c) Below is a snippet of code, explain what each line of code in this program.

```
#include <stdio.h>

int main(void)
{
    puts ("It's me, your first program.");
        return 0;
}
```

(10 marks)

d) Write a simple program that prints the name 'Wina Bwangu' 3 times (2 marks)

e) Write simple programs using 'Addition', 'Subtraction' and 'Division 'operators (12

marks)

f) Describe what the Remainder Operator is and a Literal in C programming (4 marks)

(40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

Algorithms are used to describe structured, semi-formal processes for each step in a program. This term has roots in the Arabic language and dates to the early middle Ages, which may be used as a justification to point out that computer programming dates to a very long time ago. Below is a simple program. Identify all conceivable logical and compilation errors and fix the code. Rewrite the correct code.

a)

```
#include <stdio.h>
 int main()
{
  printf("The value of six is: %d \n", 16,0-10-);
    return 0;
}
```

(10 marks)

b) What is the expected output for the snippet of code in question (2a)? (2 marks)

c) You have a string with a sentence in it. Create a C program to determine the sentence's word count. (8 marks)

(20 marks)

Question 3

An operator is a symbol of the programming language that is able to operate on the values.Operators are used to perform operations on variables and values.

Below is a snippet of code what is the value the **var** variable after the execution of the code.

a)

```
int   var;
var = 100;
var = var + 100;
var = var + var;
```

(5 marks)

b) What is the value of the **f** variable after execution of the following snippet?

```
int   i;
float f;

i = 10 / 3
f = i  * 3.0;
```

(5 marks)

c) Describe the role of the compiler in C programming language (2 marks)

d) Given the snippet below, look for any logical and compilation errors that might exist. Fix them, but don't alter any of the numbers.

```
#include <stdio.h>

int main()
{
  float halfValue = 0.6 + 0.1;

  float piValue = 0.14159265 - 3 +;

 printf("The value of half is: %f\n", halfValue);

 printf("The value of Pi is: %f\n", piValue);
    return 0;
     }
```

(6 marks)

e) Write the expected output to question (3d). (2 marks)

(20 marks)

Question 4

The following if -else execution works like this

• If the condition is "true" (that is, if its value is not zero), then perform_if_condition_true is carried out, and the conditional statement ends;

• If the condition is "false" (has a value of zero), then perform_if_condition_false function is called, ending the conditional statement;

a) Look for any logical and compilation mistakes that might exist. Fix them and find proper conditions for each of the three statements. Write the expected output for the code below.

```
#include <stdio.h>

int main(void)

{

int a = 10;

if (a  9)

puts("First condition is true");

else

puts("First condition is false");

if (a  9)

puts("Second condition is true");

else

puts("Second condition is false");

if (a is equal 9 + 1)

puts("Third condition is true");

else

puts("Third condition is false");
```

```
return 0;
}
```

(9 marks)

b) Explain the task of the '**Printf** 'and '**Scanf** ' functions in C language.State and similarities or differences in the way they function. (4 marks)

c) Create a program that prints the name of a specified weekday. Your program must produce the same output as planned. The program has been partly written for you complete it.

```
#include <stdio
 int main(void)
{
int dayOfWeek = 1;
  /* your code */
      return 0;
  }
```

(3 marks)

d) The C language has two types of conversions Implicit and Explicit. Explain each type briefly (4 marks)

(20 marks)

Question 5

In C, integers are a basic data type that can be used to represent both positive and negative whole numbers that don't have any fractional or decimal components. Depending on the architecture of the computer that the program is running on, integers in C are typically stored as a fixed number of bits. Below are snippets of code. Identify all logical and compilation errors. Rewrite the code as well as the expected output.

```
#include <stdio.h>

int main()
{
printf("The value of five is: %d \n", 5int);
return 0;
}
```

(5 marks)

b)

```
#include <stdio.h>

int main()
{
printf("The value of nine is: %d \n", 01x);
printf("The value of ten is: %d \n", 0x2);
return 0;
}
```

(5 marks)

c) Describe the values that can be stored in floating point variables and list the data types commonly used in C programming language (5 marks)

d) Define what a keyword is in the C programming language (2 marks)

e) Write the expected output in question (5a) and (5b) (3 marks)

(20 marks)

END OF EXAM